CONFOEDERATIO,
RESEARCH DIVISION (CRD).

NOVGOROD, VLADIMIR-SUZDAL AND THE RISE OF MUSCOVY

Population and Settlement in the Russian Core and the Middle Volga

8 November 2026 - Aust Kätzchen | austkaeztchen@confoederatio.org

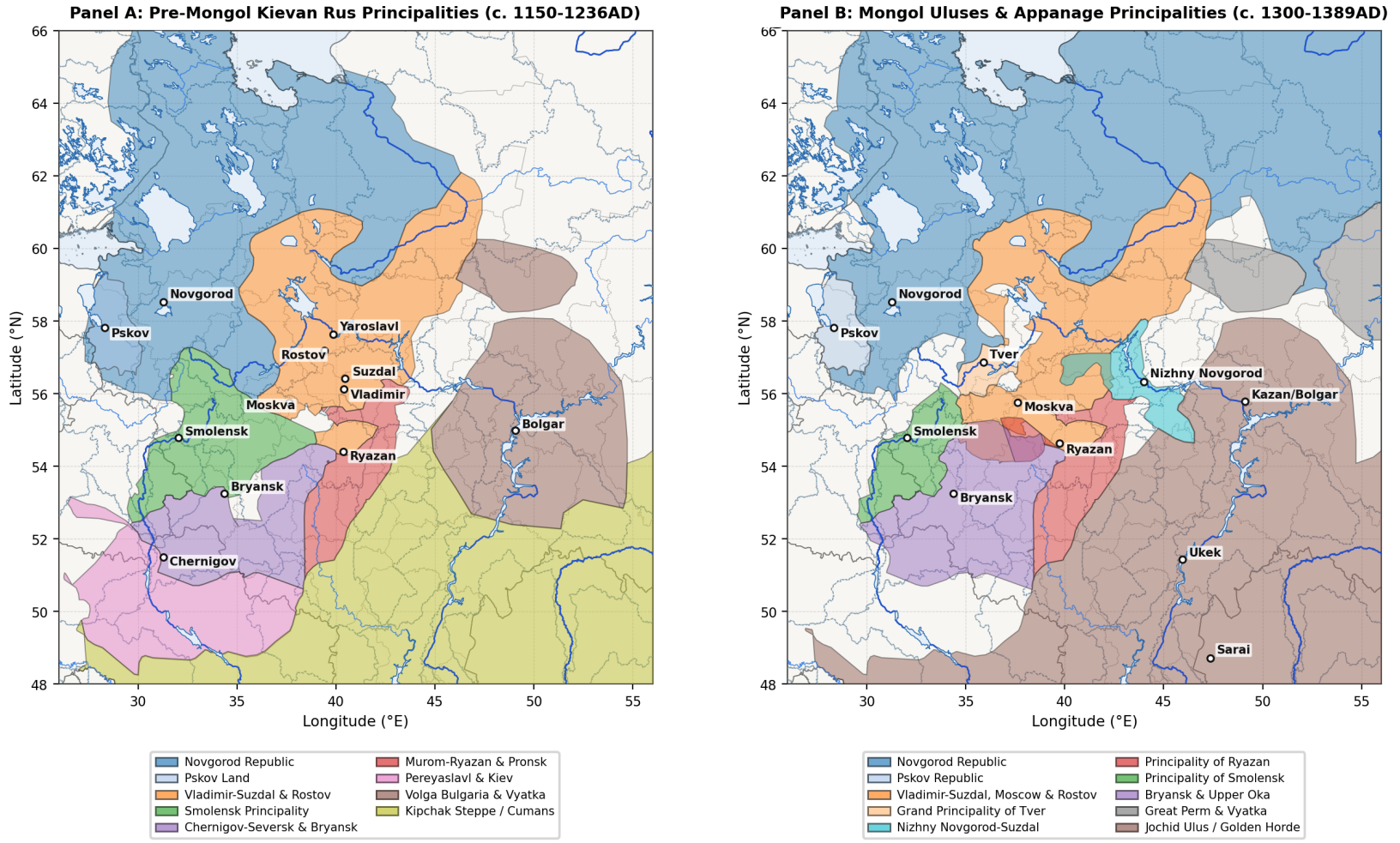


Figure 1. The eight historical regions of the Russian Core and the Middle Volga (~1,94M km^2), comparing the river basins of the twelfth-century Rus principalities and Volga Bulgaria (left) with the Golden Horde uluses and Russian appanages after 1240AD (right).

Abstract

This paper traces the population and settlement history of the Russian Core and the Middle Volga-Ural basin (~1,94M km^2) from 10000BC to 1900AD across eight historical regions: Novgorod and Pskov, Vladimir-Suzdal and Rostov-Yaroslavl, Moscow and Tver, Ryazan and the Upper Don, Smolensk and the Upper Dnieper, Chernigov-Bryansk and the Upper Oka, Nizhny Novgorod, Vyatka and Great Perm, and the Middle Volga (Volga Bulgaria and Kazan). The estimates draw on excavated house floors and hillforts for the prehistoric and Early Iron Age periods, town homestead (*usadba*) and village (*selishche*) surveys for

Kievan Rus and Volga Bulgaria, the 1257-1259 Mongol census (*chislo*), sixteenth- and seventeenth-century Muscovite tax and household registers (*pistsovye* and *perepisnye knigi*), the ten Imperial poll-tax revisions (1719-1858) and the 1897 census. Total population grew from about 13k hunters and fishers in 10000BC to 182k in the Bronze Age (2000BC), 610k in 1AD and 4,67M on the eve of the Mongol invasion (1200AD), before dropping to 3,85M by 1250AD. Clearing of the upland forests around Moscow and Tver, seventeenth-century peasant migration south into the forest-steppe and nineteenth-century farming and industrial growth then carried the total to 6,49M in 1500AD, 10,19M in 1700AD and 41,5M by 1900AD.

Background

To keep a fixed geographical frame across the prehistoric, Kievan Rus (882 - 1240AD), Mongol and Appanage (1240 - 1500AD), Muscovite (1500 - 1700AD) and Imperial (1700 - 1900AD) periods, the historical principalities are grouped into eight regions defined by river basins and later provincial borders (Figure 1 and Table 1). Because several large medieval states crossed modern administrative lines - notably the Golden Horde, the Novgorod Republic and the Chernigov-Seversk lands - the area studied here covers the Russian Core and the Middle Volga-Ural basin (~1,94M km^2), leaving out the Lower Volga steppe south of Ukek (51,5°N), the Ukrainian Dnieper-Desna left bank south of the Seym-Bryansk watershed (52°N) and Belarusian Polotsk, Turov and Vitebsk west of the Smolensk gate (31°E).

Methodology

Tracing population across the East European forests, the forest-steppe and the Middle Volga-Ural basin from 10000BC to 1900AD means joining archaeological site surveys for the prehistoric and medieval centuries to tax rolls and censuses from the sixteenth century onwards [1][2][3]. Rather than relying on rough continental totals or treating raw site counts and radiocarbon dates as direct measures of population, this study builds its figures from five main lines of evidence.

For the prehistoric and Early Iron Age centuries (10000BC - 850AD), regional densities are worked out from excavated house floors, hearth counts, cemeteries and the spacing of settlements along rivers and lakes. These include Mesolithic peat-bog camps of the Butovo, Kunda and Veretye traditions [4][5][6], timber longhouses (60 - 200 m^2, holding 15 - 40 people) of the Neolithic Lyalovo and Eneolithic Volosovo cultures [7][8], Bronze Age Fatyanovo, Abashevo and Pozdnyakovo farming and herding settlements [9][10][11] and Early Iron Age hillfort chains (*gorodishcha*, typically 0,15 - 0,5 ha set every 3 - 7 km along river bluffs, with 50 - 150 residents per fort) of the Dyakovo, Gorodets, Dnieper-Dvina and Yukhnovo cultures [12][13][14].

For the towns of Kievan Rus and Volga Bulgaria between 850 and 1450AD, estimates follow the walled-area and homestead (*usadba* or *dvor*) methods worked out by M. N. Tikhomirov [15], A. V. Kuza [2][16], V. P. Darkevich [17], V. L. Yanin [18] and F. Sh. Khuzin [19]. Across tree-ring dated quarters in Veliky Novgorod (the Nerevsky, Lyudin and Slavensky ends), Pskov, Staraya Russa, Smolensk, Old Ryazan, Vladimir and Suzdal, a town plot took up 400 - 1,2k m^2 (averaging 600 - 800 m^2 inside the timber ramparts and 900 - 1,4k m^2 in the outer *posad* suburbs), giving 10 - 16 yards per net residential hectare. At 5,5 - 6,5 people per household (or 8 - 10 on large boyar yards with dependent craftsmen), walled towns held 80 - 150 inhabitants per hectare [2][17].

Rural population figures between 900 and 1500AD come from field-walking and magnetometry surveys of unfortified villages (*selishcha*) in the Suzdal Opolye [20][21][22], the Lake Ilmen and Volkhov basin [23], the Moskva-Radonezh uplands [24][25], the Upper Don valley [26] and the Western Trans-Kama (*Zakamye*) farmlands of Volga Bulgaria [27][28]. Whereas a small northern forest hamlet (*derevnya* of 1 - 3 yards) covered 0,3 - 1,2 ha (15 - 30 inhabitants), large Opolye and Bulgar black-earth villages (*sela*) covered 5 - 19 ha and housed 100 - 350 people [19][21].

For the Mongol and early modern periods between 1257 and 1700AD, estimates are checked against the Mongol imperial census (*chislo*) taken under Möngke and Berke in 1257 - 1259 [29][30][31], the late fifteenth- and sixteenth-century Novgorodian and Muscovite land cadastres (*pistsovye knigi*, which counted peasant *dvory* and *obzha* or *sokha* plough units) [32][33][34] and the 1646 and 1678 household census books (*perepisnye knigi*) studied by Ya. E. Vodarsky [1][3][35]. From Peter I's reign to the end of the nineteenth century (1719 - 1900AD), provincial totals are anchored to the ten Imperial poll-tax revisions (*revizskie skazki*, 1719 - 1858) compiled by V. M. Kabuzan [36][37][38][39], nineteenth-century provincial statistics gathered by A. G. Rashin and B. N. Mironov [40][41][42][43] and the First All-Russian Imperial Census of 28 January 1897 edited by N. A. Troynitsky [44][45], carried forward to 1900AD.

Regional Division	Constituent Medieval Polities	Kievan Span	Mongol / Appanage	Corresponding Imperial Provinces & Area
Novgorod, Pskov, St. Petersburg & N. Dvina	Novgorod Republic, Pskov Republic, Zavolochye	882 - 1240AD	1240 - 1478/ 1510AD	St. Petersburg, Novgorod, Pskov, Olonets, Arkhangelsk (680k km^2)
Vladimir-Suzdal, Rostov-Yaroslavl & Beloozero	Vladimir-Suzdal, Rostov, Yaroslavl, Beloozero	988 - 1238AD	1238 - 1474AD	Vladimir, Yaroslavl, Kostroma, Vologda (285k km^2)
Moscow & Tver	Grand Principalities of Moscow and Tver	1147 - 1238AD	1247 - 1485AD	Moscow, Tver (92k km^2)
Ryazan, Murom, Pronsk & Upper Don	Murom-Ryazan, Pronsk, Yelets, Lipetsk	1097 - 1237AD	1237 - 1521AD	Ryazan, N. Tambov, N. Voronezh / Upper Don (105k km^2)
Smolensk, Toropets & Upper Dnieper	Smolensk, Toropets, Vyazma, Dorogobuzh, Mstislavl	1054 - 1239AD	1239 - 1404/ 1514AD	Smolensk, E. Toropets / Bely (78k km^2)
Chernigov-Bryansk, Kursk & Upper Oka	Chernigov-Kursk, Bryansk, Upper Oka Principalities	1024 - 1239AD	1239 - 1503AD	Kursk, Orel, Tula, Kaluga, Bryansk (108k km^2)
Nizhny Novgorod, Vyatka & Great Perm	Nizhny Novgorod-Gorodets, Vyatka Land, Great Perm	1152 - 1239AD	1239 - 1489/ 1505AD	Nizhny Novgorod, Vyatka, Perm (420k km^2)
Middle Volga (Bulgar, Kazan, Mukhshi, Kasimov)	Volga Bulgaria, Bulgar and Mukhshi Uluses, Kazan, Kasimov	922 - 1236AD	1236 - 1552AD	Kazan, Simbirsk, Penza, Kasimov/ Temnikov (175k km^2)

Table 1. The eight regional divisions of the Russian Core, matching the medieval principalities and Volga khanates to their Kievan and appanage date spans and the nineteenth-century imperial guberniyas used to align census totals.

Results

Prehistoric and Early Iron Age Settlement

Mesolithic and Pit-Comb Neolithic Foragers and Fishers

Human settlement returned to the East European forest zone at the end of the Ice Age (10000 - 8000BC) as mobile bands of reindeer and elk hunters followed the river valleys north [4][5]. Between the Volga and Oka and across the Upper Dnieper-Desna basin, Butovo and Resseta hunters (8000 - 5500BC) pitched seasonal camps on sandy lake terraces and peat bogs such as Stanovoye 4, Ivanovskoye 7 and Zamostje 2 [5][6]. In the north-western lake country around Lakes Ilmen, Lacha, Vozhe and Kubenskoye and along the Sukhona, Kunda and Veretye hunter-fishers built log-cabin lakeside villages such as Nizhneye Veretye I (6800 - 6000BC), where waterlogged peat has preserved bone harpoons, fish weirs and birch-bark floats [4]. Because food supplies depended on seasonal deer runs and lake fishing, densities stayed very low (0,006 - 0,025 persons/km^2), and the population of the entire Russian Core and Middle Volga-Ural territory rose slowly from about 13k in 10000BC to 44,5k by 6000BC [6].

During the warmer Atlantic period (5500 - 2800BC), Upper Volga early pottery groups (5500 - 4200BC), the Lyalovo pit-comb ware culture (4200 - 3000BC) and the Eneolithic Volosovo culture (3200 - 2000BC) supported a much thicker scatter of year-round fishing and hunting villages at lake outlets across the Volga-Oka, Klyazma and Kama basins [7][8]. At the Sakhtysh I-VIII peat-bog sites near Ivanovo and on Lakes Nero and Pleshcheyevo, Volosovo villagers built sunken timber longhouses of 60 - 200 m^2 with 3 - 6 hearths down the centre, each housing 15 - 40 people alongside cemeteries of more than 100 graves [7][8][10]. Across all eight regions, total numbers climbed from 86k in 4000BC to 122k by 3000BC.

Bronze Age Herders and Iron Age Hillforts

Between 2800 and 2000BC, Corded Ware settlers of the Fatyanovo-Balanovo culture brought domestic cattle, sheep, goats, pigs and wheeled carts into the Upper and Middle Volga forests [9][10]. Ancient DNA from Fatyanovo cemeteries in Yaroslavl, Ivanovo, Moscow and Chuvashia (including Volosovo-Danilovsky, with over 100 graves) shows that these newcomers combined steppe ancestry with Central European farming stock and lived alongside native Volosovo hunter-fishers for several centuries before merging with them [9][10]. Later Middle and Late Bronze Age groups - the Abashevo (2200 - 1700BC), Seima-Turbino (2100 - 1800BC), Pozdnyakovo (1800 - 1200BC) and Prikazan (1600 - 900BC) cultures - spread copper and bronze working along the Oka, Klyazma, Kostroma and Kama rivers, as seen at the Starshe-Nikitinsky cemetery and the Govyadinovo settlement [11][46]. Total population reached 182k in 2000BC and 278k by 1000BC.

The Early Iron Age (800BC - 500AD) brought local bog-iron smelting, horse, cattle and pig rearing and slash-and-burn millet and barley farming [12][13][14]. Across the Upper Volga, Moskva and Oka basins, the Finno-Ugric Dyakovo culture (700BC - 500AD) built hundreds of bluff-top hillforts (*gorodishcha*) ringed by ditches and log stockades, while the related Gorodets culture held the Middle Oka and Moksha-Sura forest-steppe, the Dnieper-Dvina and Yukhnovo cultures occupied the Smolensk and Bryansk-Desna valleys and the Ananyino (800 - 300BC), Pyanobor (200BC - 300AD) and Glyadenovo cultures settled the Kama and Vyatka valleys [12][47]. N. A. Krenke's excavations at Dyakovo Gorodishche, Kuntsevo and Borsheva on the Moskva river show that these hillforts enclosed 0,15 - 0,5 ha, stood 3 - 7 km apart along the main rivers and held 50 - 150 people (8 - 20 timber houses), each surrounded by small open hamlets (*selishcha*) [12][13][48]. By 1AD the eight regions held 610k inhabitants, rising to 920k in 500AD and 1,39M by 800AD as Early Slavic settlers (Ilmen Slovenes, Krivichi, Vyatichi and Severians) and Early Bulgars moved into the forest and forest-steppe belts [49][50].

Kievan Rus Principalities

Novgorod and Pskov

Between the ninth and eleventh centuries, the north-western lake basin grew rich on the river trade linking the Baltic to the Volga and the Black Sea [23][49]. Early trading and craft settlements at Staraya Ladoga (c. 753AD), Rurikovo Gorodishche at the Volkhov outlet (ninth to tenth centuries, 4 - 7 ha) and the high-mound (*sopki*) villages around Lake Ilmen gave way in the mid-tenth century to the three founding quarters (*kontsy*: Nerevsky, Slavensky and Lyudin) of Veliky Novgorod [18][23]. By 1200AD the outer ramparts (*Okolny Gorod*) of Veliky Novgorod enclosed about 240 ha and 4k - 5k fenced courtyard homesteads (*usadby* of 600 - 1,2k m^2 excavated in the Troitsky and Nerevsky trenches), giving the city 25k - 30k inhabitants [2][15][18].

To the west, in the Lake Peipus and Velikaya basin, Pskov grew from its 3 ha Krom promontory fort (eighth to tenth centuries) across the Dovmontov Gorod and the Middle and Outer Town walls (~105 ha by the fourteenth century) to 12k - 16k residents, backed by smaller fortresses at Izborsk, Ostrov, Opochka and Gdov [51][52]. Other major eleventh- and twelfth-century Novgorodian towns included Staraya Russa (8k - 10k, the main salt-boiling town south of Lake Ilmen), Torzhok (5k - 7k on the Tvertsa portage), Ladoga (3k - 4k), Velikiye Luki and Koporye [2]. Across the farmland of the Five Pyatinas (Vodskaya, Obonezhskaya, Bezhetskaya, Derevskaya and Shelonskaya) and the Northern Dvina (*Zavolochye*) fur-trapping settlements, the population of the Novgorod-Pskov region climbed from 285k in 900AD to 380k in 1000AD and 620k on the eve of the Mongol invasion (1200AD) [32].

Vladimir-Suzdal, Rostov-Yaroslavl and Beloozero

In North-Eastern Rus, Slavic farmers from the Krivichi and Ilmen Slovene lands mingled with the native Finno-Ugric Merya and Ves between the tenth and twelfth centuries [22][53][54]. The richest farmland lay in the Suzdal Opolye (~4,5k km^2 of well-drained dark loess soil between Suzdal, Vladimir, Yuriev-Polsky and Pereyaslavl-Zalessky). Over twenty years of field-walking and magnetometry led by N. A. Makarov, I. N. Modin and S. V. Shpolyanskiy have mapped nearly 400 medieval villages (*selishcha*) in the Opolye alone [20][21]. Roughly a quarter to a third of twelfth- and early thirteenth-century Opolye villages (such as Shekshovo-2/9, Gnezdilovo, Kideksha and Ves) were large multi-yard settlements of 5 - 19 ha (100 - 350 inhabitants each), giving rural densities of 18 - 28 persons/km^2 across the Opolye soils [20][21].

Under Princes Yuri Dolgoruky (1125 - 1157), Andrey Bogolyubsky (1157 - 1174) and Vsevolod the Big Nest (1176 - 1212), a chain of fortified towns sprang up across the Klyazma and Upper Volga basin (Figure 2) [54][55]. By 1200 - 1236AD the capital at Vladimir-on-Klyazma enclosed 145 ha within its Pecherny Gorod, Novy Gorod and Vetchany Gorod ramparts and held 20k - 25k people; Rostov Veliky (65 ha), Suzdal (49 ha) and Pereyaslavl-Zalessky (42 ha) each housed 8k - 12k; and Yaroslavl, Yuriev-Polsky, Kostroma, Galich-Mersky, Starodub-on-Klyazma, Gorodets-Radilov (60 ha), Uglich and Dmitrov had 3k - 7k apiece [2][16][56]. Further north on Lake Beloye and the Sheksna river, Beloozero anchored a string of Finno-Slavic settlements surveyed by A. V. Kudryashov [57]. Altogether, the Vladimir-Suzdal, Rostov-Yaroslavl and Beloozero region grew from 310k in 900AD and 430k in 1000AD to 920k by 1200AD [56].

Smolensk, Chernigov-Bryansk, the Upper Oka and Murom-Ryazan

Along the portages between the Upper Dnieper and the Western Dvina, tenth-century town life centred on Gnezdovo (12 km west of Smolensk), where two forts (17 - 20 ha of occupation) and more than 4,5k burial mounds (*kurgany*) point to a permanent population of 1,5k - 2,5k Scandinavian, Slavic and Baltic residents [58][59]. In the eleventh and twelfth centuries princely rule moved to Smolensk, whose 115 ha walled town and crowded riverside craft quarters (*posad*) held 20k - 30k people before the famine and

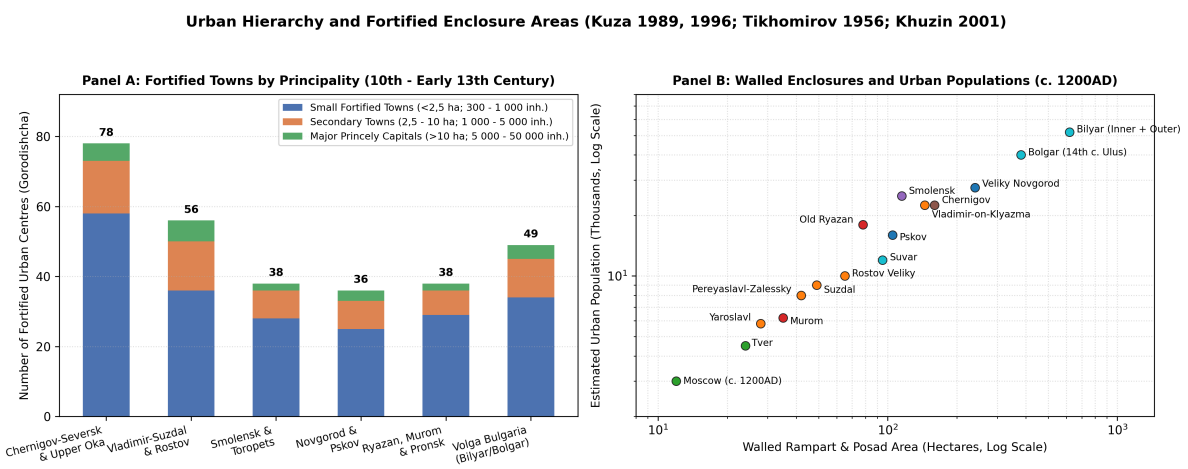


Figure 2. Fortified towns across the Russian Core and Volga Bulgaria around 1200AD, comparing the distribution of small fortresses (<2,5 ha), middle-rank towns and major capitals by region (left) with the walled area and population of the largest cities from Bilyar (620 ha, 45k - 60k) and Veliky Novgorod (240 ha, 25k - 30k) down to Murom (right).

plague of 1230, supported by provincial towns at Toropets, Roslavl, Mstislavl, Dorogobuzh, Vyazma and Yelnya [2][15][60]. Across the 56 tax districts (*pogosty*) listed in Prince Rostislav Mstislavich's charter of 1136, the Smolensk principality grew from 210k in 1000AD to 430k in 1200AD [60].

South and east of Smolensk, the Russian part of the Chernigov-Seversk and Upper Oka lands covered the Vyatichi territory along the Upper Oka, Zhizdra, Upa and Zusha rivers as well as the Desna-Bryansk and Posemye (Kursk-Rylsk) valleys [50][61]. As Figure 2 shows, A. V. Kuza's survey of tenth- to thirteenth-century fortified sites (*gorodishcha*) found that the Chernigov-Seversk and Upper Oka country had one of the thickest clusters of small fortresses and refuge towns (<2,5 ha) in Rus, including Kozelsk, Karachev, Bryansk (*Debryansk*), Mtsensk, Novosil, Tarusa, Vorotynsk, Odoev, Belyov, Mezetsk, Mosalsk, Serensk (known for its twelfth-century jewellery moulds), Dedoslavl, Spash-Gorodok and Kursk [2][16][61]. Taken together with the larger towns of the Desna-Seym basin, the Russian sector of Chernigov-Bryansk and the Upper Oka grew from 310k in 1000AD to 580k in 1200AD.

In the Middle Oka and Upper Don forest-steppe, the Murom-Ryazan and Pronsk principalities filled out rapidly during the twelfth century [17]. Excavations by V. P. Darkevich and G. V. Borisevich at Old Ryazan (*Staraya Ryazan*, near Spassk-Ryazansky) show that its Northern (8 ha), Southern (60 ha) and outer *posad* enclosures (~78 ha in all) held 1,5k - 2,5k timber yards and workshops, housing 15k - 20k people before Batu Khan burned the city in December 1237 [2][17]. Together with Murom (35 ha, 6k - 8k), Pronsk (4k - 6k), Pereyaslavl-Ryazansky, Kolomna, Rostislavl and Belgorod-Ryazansky, and more than 250 twelfth- to fourteenth-century rural villages surveyed along the Upper Don and Nepryadva [26], the Ryazan, Murom and Upper Don region reached 520k inhabitants by 1200AD (Table 2).

Urban Centre	Region	Walled & Posad Area (ha)	Estimated Homesteads	Estimated Population
Bilyar (Volga Bulgaria)	Middle Volga	620 (800 w/ suburbs)	7,5k - 9,5k	45k - 60k (c. 1200)
Bolgar (Golden Horde Ulus)	Middle Volga	380 (14th c. rampart)	5,5k - 7k	35k - 45k (c. 1340)
Veliky Novgorod	Novgorod and Pskov	240 (Okolny Gorod)	4,2k - 5k	25k - 30k (c. 1200)

Smolensk	Smolensk and Upper Dnieper	115 (Detinets + Posad)	3,5k - 4,5k	22k - 28k (c. 1200)
Vladimir-on-Klyazma	Vladimir-Suzdal	145 (3 rampart rings)	3,2k - 4k	20k - 25k (c. 1200)
Chernigov (Desna Core)	Chernigov-Bryansk	160 (Detinets + Okolny)	3,2k - 4k	20k - 25k (c. 1200)
Old Ryazan (Staraya Ryazan)	Ryazan and Upper Don	78 (North + South + Posad)	2,2k - 3k	15k - 20k (c. 1236)
Pskov	Novgorod and Pskov	105 (Krom + Outer Wall)	2,2k - 2,8k	14k - 18k (c. 1350)
Suvar (Volga Bulgaria)	Middle Volga	95 (Walled Shahristan)	1,8k - 2,2k	10k - 14k (c. 1200)
Rostov Veliky	Rostov-Yaroslavl	65 (Rampart enclosure)	1,5k - 1,9k	9k - 12k (c. 1200)
Suzdal	Vladimir-Suzdal	49 (Kremlin + Okolny)	1,3k - 1,6k	8k - 10k (c. 1200)
Pereyaslavl-Zalessky	Vladimir-Suzdal	42 (1152 Earth Rampart)	1,1k - 1,4k	7k - 9k (c. 1200)
Staraya Russa	Novgorod and Pskov	38 (Saline Posad)	1,2k - 1,5k	8k - 10k (c. 1200)
Murom	Ryazan and Murom	35 (Kremlin + Posad)	950 - 1,2k	6k - 7,5k (c. 1200)
Yaroslavl	Rostov-Yaroslavl	28 (13th c. Ramparts)	850 - 1,1k	5k - 7k (c. 1200)
Tver	Moscow and Tver	24 (1200)/85 (1400)	750 / 2,2k	4,5k (1200)/14k (1400)
Moscow (1200/ 1450AD)	Moscow and Tver	12 (1200)/190 (1450)	450 / 6,5k	3k (1200)/40k (1450)

Table 2. Major medieval cities of the Russian Core and Volga Bulgaria (1200 - 1450AD), showing how walled and suburban areas in hectares translate into homestead counts (at 10 - 16 yards/ha) and urban populations before and after the Mongol conquest.

Mongol Census, Appanages and Volga Khanates

The 1237-1240 Invasion, the 1257-1259 Census and Appanage Settlement

Batu Khan's winter campaign of 1237 - 1238 against Ryazan, Kolomna, Moscow, Vladimir, Suzdal, Rostov, Yaroslavl, Tver, Torzhok and Kozelsk - preceded by Subutai's conquest of Volga Bulgaria in 1236 and followed by the 1239 attacks on Murom, Gorodets and Chernigov - hit the walled towns and exposed forest-steppe valleys hard [31][62][63]. Out of 74 Russian towns known from twelfth- and early thirteenth-century excavations, 49 were burned between 1237 and 1240 and 14 (including Old Ryazan and Izheslavl) never recovered as towns [2][17]. Between 1200 and 1250AD total population across the eight regions fell by about 18%, from 4,67M to 3,85M, with the sharpest drops in Ryazan (-28,8%), Chernigov and the Upper Oka (-25,9%), Volga Bulgaria (-22,4%) and Vladimir-Suzdal (-19,6%), whereas uninvaded Novgorod and Pskov lost less than 5% [56][62].

To organise tribute (*vykhod* or *dan'*) and military levies across the Jochid realm, Grand Qan Möngke and Berke sent census officials (*chislenniki*) in 1257 - 1259 to count households in Suzdalia, Ryazan, Murom and (backed by Alexander Nevsky's troops in 1259) Veliky Novgorod [3][29][31]. As G. Vernadsky [30] and T. T.

Allsen [29] showed, the Mongol scribes grouped tax-paying peasant households into decimal units of ten, a hundred, a thousand (*tysyacha*) and ten thousand (*tumen* or *t'ma*). The 27 to 43 *t'my* mentioned in fourteenth-century princely treaties and chronicles across North-Eastern Rus and Novgorod - where a tax *t'ma* in the forest zone answered to 10k - 15k peasant households (60k - 90k people) rather than a full corps of 10k horsemen - fit an overall post-invasion population of 3,85M in 1250AD, recovering to 4,33M in 1300AD and 4,84M on the eve of the Black Death (1350AD) [29][30][54].

Between the late thirteenth and fifteenth centuries, rural surveys show a major shift in peasant settlement across Moscow, Tver and Vladimir-Suzdal known to Russian historians as the "move to the hills" (*vzlet na kholmy*) or watershed clearing [21][24][25]. Instead of staying crowded along the main river floodplains where they were exposed to Golden Horde punitive raids (*rati* such as Dyuden's march of 1293 or Tokhtamysh's sack of 1382), peasants cleared the higher clay ridges between the rivers with iron-shod *sokha* ploughs and three-field rotation (winter rye, spring oats and fallow). S. Z. Chernov's complete walks of the Radonezh, Zvenigorod, Vorya and Pereyaslavl rural districts (*volosti*) show that between 1280 and 1450AD hundreds of small 1 - 3 household clearings (*pochinki* and *derevni*, 0,2 - 0,8 ha) spread across the upland woods under the protection of the Grand Princes of Moscow and monasteries such as the Trinity-Sergius Lavra [24][25]. At the same time, the Grand Principality of Tver (with Kashin, Mikulin, Zubtsov, Kholm and Staritsa) and the Principality of Nizhny Novgorod-Suzdal (founded in 1341 where the Oka meets the Volga) grew into strong rivals to Moscow [54][64]. Although the Black Death (which reached Rus in 1352 - 1353, killing Grand Prince Simeon the Proud and Metropolitan Theognostus, and returned in 1364 - 1366, 1389 and 1417 - 1427) cut the total population back from 4,84M in 1350AD to 4,49M in 1400AD, steady forest clearing lifted the Moscow-Tver region from 330k in 1250AD to 830k in 1450AD and 1,08M by 1500AD [33][65].

Volga Bulgaria, the Golden Horde Uluses and the Kazan, Kasimov and Temnikov Lands

In the Middle Volga and Kama basin, pre-Mongol Volga Bulgaria (922 - 1236AD) was one of the most heavily urbanised states in Eastern Europe [19][27]. F. Sh. Khuzin's and K. A. Rudenko's surveys of more than 170 Bulgar walled towns (*gorodishcha*) and over 700 open villages (*selishcha*) show that the twelfth-century capital of Bilyar (*Velikiy Gorod* in Russian chronicles) covered 620 ha inside its triple earth ramparts (800 ha with its outer suburbs), with brick-heated houses, glass and metal workshops and a peak population of 45k - 60k [19][27]. Other leading Bulgar towns included Suvar (95 ha, 10k - 14k), Bolgar (25 - 30 ha in the twelfth century), Oshel, Juketau (*Zhukotin*), Kashan, Muromsky Gorodok (150 ha on the Samara Bend) and Kazan (founded c. 1005AD) [19]. Across the fertile black-earth plains of Western Trans-Kama (*Zakamye*, along the Cheremshan, Sheshma, Aktay and Maly Cheremshan rivers), pre-Mongol Volga Bulgaria held 850k people by 1200AD [27].

After the Mongol conquest of 1236 destroyed Bilyar, Oshel and Suvar, the territory was reorganised as the Bulgar Ulus and the southern Ulus of Mukhshi (Narovchat) within the Golden Horde [28][66][67]. Under Mengu-Timur and Uzbeg Khan (1313 - 1341), Bolgar grew into a 380 ha walled city of 35k - 45k inhabitants that served as the northern summer residence and mint of the Jochid khans [28], while in the Moksha-Sura forest-steppe (modern Penza Oblast and Mordovia) Mukhshi (Narovchat, over 150 ha, with brick mosques, mausoleums, baths and a silver mint opened in 1313) stood at the centre of a thick cluster of Mordvin, Burtas and Tatar villages [66][67]. East of the Kama, between the Ik and Belaya rivers, Tatar-Muslim farming and herding settlements also spread during the thirteenth and fourteenth centuries [68].

During the late fourteenth century, however, the Golden Horde's civil wars (*Velikaya Zamyatnya*, 1359 - 1380), Bulat-Timur's seizure of Bolgar in 1361, Muscovite river raids (1376, 1399), Timur's campaign of 1395 and Nogai raids drove farmers north from the exposed Southern Zakamye steppe border into the wooded Predkamye north of the Kama (along the Kazanka, Mesha, Ashit, Ilet and Sviyaga rivers) [27][34]. This northward shift formed the heartland of the Khanate of Kazan (1438 - 1552AD), divided into five

administrative districts (*darugas*: Alat, Arsk, Galich, Zyuri and Nogai) and inhabited by Kazan Tatars, Chuvash, Mari (Cheremis), Udmurts (Votyaks) and Mordvins (860k across the Middle Volga region in 1500AD) [34][69]. On the Oka-Moksha border between Ryazan and Kazan, the Kasimov Khanate (founded c. 1452 at Gorodets-Meshchersky) and the Temnikov Principality (where Tatar princes of the Bekhanid line ruled Mordvin *belyaki* tribute districts) formed a settled military frontier recorded in sixteenth- and seventeenth-century service rolls [70][71]. Meanwhile, along the Vyatka and Upper Kama rivers, Russian settlers from Novgorod, Ustyug and Galich built up the self-governing Vyatka Land (around Khlynov, Kotelnich and Orlov, alongside the Arsk and Karino Tatar princes on the Cheptsa river; annexed by Ivan III in 1489) and the Principality of Great Perm (around Cherdyn, Pokcha and Iskor where the Kolva and Vishera join the Kama, growing out of the medieval Komi-Permyak Lomovatovo and Rodanovo cultures; brought under Moscow in 1472 - 1505) [47][72][73][74][75].

Muscovite Cadastral Registers and the 1678 Household Census

Once Ivan III (1462 - 1505) and Vasily III (1505 - 1533) had brought the northern principalities under Moscow's rule, Muscovite scribes drew up land cadastres (*pistsovye knigi*) listing towns and rural parishes (*pogosty*) household by household [3][32][33]. In the Novgorod Five Pyatinas (1495 - 1505), A. L. Shapiro and A. I. Kopanev's study of the surviving books counts 65k - 70k peasant homesteads (*dvory*) rated in *obzha* plough units (1 *obzha* = 1 man with 1 horse ploughing ~15 *desyatiny* across three fields), pointing to 500k - 550k people in the Five Pyatinas proper and 810k across the whole Novgorod, Pskov and Northern Dvina region in 1500AD, rising to 940k by 1550AD [32][33]. Across the Zamoskovny heartland (Moscow, Tver, Vladimir, Suzdal, Pereyaslavl, Rostov, Yaroslavl, Kostroma and Beloozero), early sixteenth-century cadastres indicate 2,57M inhabitants in 1500AD (1,49M in Vladimir-Suzdal and Rostov-Yaroslavl and 1,08M in Moscow-Tver), growing to 3,09M by 1550AD as three-field ploughland reached its widest pre-industrial extent in the northern forest belt [1][33]. In the newly conquered Kazan district (*uezd*), the 1565 - 1568 cadastres of Nikita Borisov and Dmitry Kikin list more than 700 Russian, Tatar and Chuvash villages across the Kazanka, Sviyaga and Kama *darugas* [34][69].

In the late sixteenth and early seventeenth centuries the northern and western provinces were set back by two great crises: first, the Livonian War (1558 - 1583), Ivan IV's Oprichnina terror (1565 - 1572, including the sack of Novgorod in 1570) and the famine and plague of 1569 - 1571, which left 60 - 85% of rated plough units (*obzhi*) in the Novgorod Pyatinas and Moscow district lying waste (*pustoshi*) in the 1580s tax books [32][76]; and second, the Time of Troubles (*Smuta*, 1598 - 1613), touched off by the famine of 1601 - 1603 and Polish-Lithuanian and Swedish invasions [1][65]. Across all eight regions, total population fell from 7,65M in 1550AD to 6,75M in 1600AD.

Recovery in the seventeenth century went hand in hand with a steady movement of peasants south behind the Belgorod and Simbirsk fortified lines (*Zasechnaya cherta*, built in 1635 - 1654), recorded in the 1646 and 1678 household census books (*perepisnye knigi*) studied by Ya. E. Vodarsky [1][3][35]. Because seventeenth-century peasant families often kept married brothers or grown sons inside a single fenced yard to avoid paying tax on a second household (*dvor*), Vodarsky showed that the average rural household in 1678 held 6,5 - 8,5 people (3,5 - 4,2 males per *dvor*). As Figure 3 shows, between 1500 and 1700AD the population of the old north-western forest country barely passed its mid-sixteenth-century peak (reaching 1,01M in 1700AD), whereas the forest-steppe provinces along the Upper Oka, Kursk, Orel and Tula more than doubled from 650k to 1,35M, the Ryazan and Upper Don black-earth frontier climbed from 560k to 1,05M, the Middle Volga rose from 860k to 1,18M and the Vyatka-Kama-Perm lands grew from 610k to 915k, bringing the total across the eight regions to 10,19M by 1700AD [35].

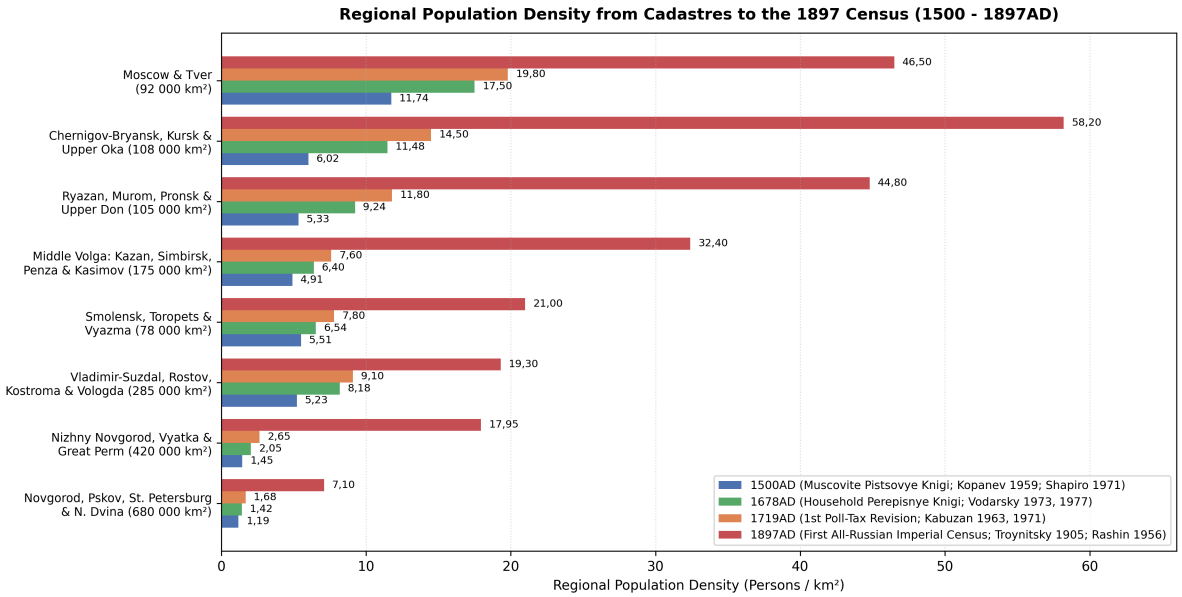


Figure 3. Shift in population density (inhabitants/km^2) from the northern forest heartland to the southern black-earth provinces across four fiscal benchmarks: the 1500AD Muscovite cadastres, the 1678 household census, the Fifth Revision of 1795 and the 1897 imperial census.

Imperial Tax Revisions and the 1897 Census

Peter I's replacement of the household tax with a head tax on every male soul (*podushnaya podat'*) in 1718 - 1724 launched ten Imperial poll-tax revisions (*revizii* or *revizskie skazki*) between 1719 and 1858, which V. M. Kabuzan compiled and checked against parish birth and death registers [36][37][38][39], followed by late nineteenth-century zemstvo counts and the First All-Russian Imperial Census of 28 January 1897 [40][41][44]. Over the two hundred years from 1700 to 1900AD, the combined population of the eight regions quadrupled from 10,19M to 41,5M (Figure 4).

In the north-west, Peter I's founding of Saint Petersburg on the Neva delta in 1703 (inside the old Ingrian and Vodskaya Pyatina lands of Novgorod) created a fast-growing capital and seaport that reached ~95k people by 1750, 220k in 1800, 490k in 1850 and 1,26M in the 1897 census (2,11M across Saint Petersburg Guberniya) [40][44]. Taken together with Novgorod (1,37M in 1897), Pskov (1,12M), Olonets (364,2k) and the Northern Dvina and Arkhangelsk districts, the north-western region grew from 1,01M in 1700AD to 1,95M in 1800AD and 4,95M by 1900AD [37][44].

Throughout the eighteenth and early nineteenth centuries (from the First to the Tenth Revision, 1719 - 1858), the rich Central Black Earth (*Chernozem*) provinces south of the Oka had the fastest natural increase and thickest rural settlement in European Russia [37][42][43]. Across Kursk (2,37M in 1897), Orel with Bryansk and Karachev (2,03M), Tula (1,42M) and Kaluga (1,13M), population climbed from 1,35M in 1700AD to 2,95M in 1800AD and 6,45M in 1900AD (58,2 persons/km^2 in 1897) [40][44]. In the same way, Ryazan Guberniya (1,8M in 1897) and the northern Upper Don districts of Tambov and Voronezh (Lipetsk, Yelets, Lebedyan, Dankov, Usman, Shatsk and Morshansk) rose from 1,05M in 1700AD to 2,25M in 1800AD and 4,85M by 1900AD [43][44].

In the old Zamoskovny forest provinces, thin sandy and clay soils pushed peasants into seasonal craft work (*otkhodnichestvo*), linen and cotton weaving (around Moscow, Ivanovo-Voznesensk, Shuya, Orekhovo-Zuyevo, Yaroslavl and Kostroma) and factory towns along the post-1861 railways [39][41]. By the

1897 census Moscow Guberniya had 2,43M inhabitants (including 1,04M in Moscow city) and Tver had 1,77M, carrying the Moscow-Tver division to 4,42M in 1900AD; alongside it, the Vladimir-Suzdal and Rostov-Yaroslavl division - covering Vladimir (1,52M), Kostroma (1,39M), Vologda (1,34M) and Yaroslavl (1,07M) together with the Belozersk and Cherepovets districts - reached 5,65M by 1900AD [40][44]. Growth was faster still across Nizhny Novgorod (1,52M), Vyatka (3,03M) and Perm (2,99M), where state peasant clearing in Vyatka and the eighteenth- and nineteenth-century boom in Ural iron and copper works multiplied the population more than eightfold, from 915k in 1700AD to 2,42M in 1800AD and 7,65M in 1900AD [37][44]. In the Middle Volga (Kazan Guberniya at 2,17M in 1897, Simbirsk at 1,53M, Penza at 1,47M and the Kasimov, Elatma and Temnikov districts at ~680k), Russian, Tatar, Chuvash, Mordvin and Mari farming raised the population from 1,18M in 1700AD to 2,47M in 1800AD and 5,85M in 1900AD [44][45].

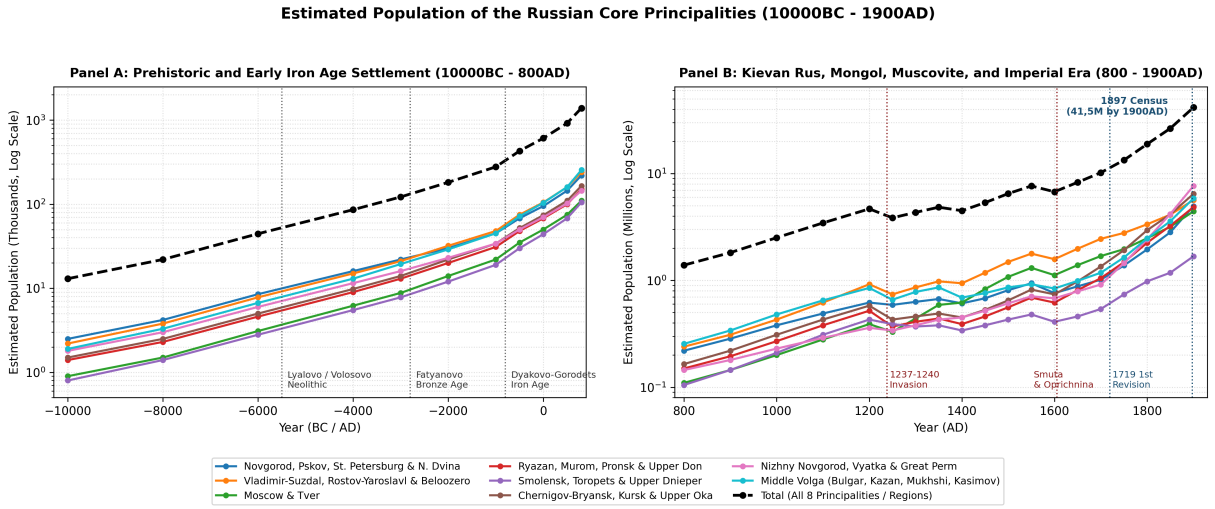


Figure 4. Long-run population growth across the eight regions of the Russian Core and the Middle Volga, contrasting the slow rise of prehistoric and medieval numbers with the effects of the 1237 - 1240 Mongol invasion, the Time of Troubles and the fourfold expansion between 1700 and 1900AD.

Limitations

Three main sources of uncertainty affect pre-census population estimates across the Russian Core and Middle Volga-Ural regions. First, in the northern forest belt, slash-and-burn clearing (*podseka*) and periodic village moves before the spread of three-field rotation in the fourteenth and fifteenth centuries mean that open rural villages (*selishcha*) were often lived in for only two to four generations, so raw survey site counts must be adjusted for how long each village stood [20][24]. Second, sixteenth- and seventeenth-century Muscovite tax books (*pistsovye* and *perepisnye knigi*) rated peasant homesteads (*dvory*) and plough units (*obzhi* and *sokhi*), which gave extended families a strong reason to crowd into shared courtyards and hide untaxed cottagers (*bobyli* and *zakhrebetniki*) until Peter I brought in the individual male head tax in 1718 - 1724 [32][35]. Third, along the Middle Volga and Kama-Ural borderlands, non-Slavic tribute payers (*yasachnye lyudi*) were only partly counted in the first two poll-tax revisions until the Third to Fifth Revisions (1762 - 1795) brought all rural classes onto the same rolls [34][37].

Conclusion

Across the eleven millennia from the end of the Ice Age to 1900AD, the population of the Russian Core and the Middle Volga-Ural basin grew from about 13k hunters and fishers to 41,5M inhabitants on the eve of the twentieth century. Pre-Mongol growth rested on the river trading towns of Novgorod, Smolensk, Vladimir-Suzdal, Ryazan and Volga Bulgaria, reaching 4,67M by 1200AD before the Mongol campaigns of 1236 - 1240 cut the total back to 3,85M. Thereafter, peasant clearing of the upland clay hills around Moscow, Tver and the Upper Volga during the fourteenth and fifteenth centuries (6,49M by 1500AD), seventeenth-century settlement south of the Oka fortified lines (10,19M by 1700AD) and eighteenth- and nineteenth-century black-earth farming and factory growth shifted the weight of Russia's population from the old northern forest principalities to the southern and eastern provinces.

Appendix

Population Estimates by Principality

Table 3 sets out the regional population estimates across all eight divisions from 10000BC to 1900AD.

Year	Novg/ SPb	Vlad-Suzd	Mosc-Tver	Ryaz-Don	Smol.	Oka-Kursk	N.Nov-Perm	Mid-Volga	Total
10000BC	2,5k	2,2k	0,9k	1,4k	0,8k	1,5k	1,8k	1,9k	13k
8000BC	4,2k	3,8k	1,5k	2,3k	1,4k	2,5k	3k	3,3k	22k
6000BC	8,5k	7,8k	3,1k	4,6k	2,8k	5k	6k	6,7k	44,5k
4000BC	16k	15k	6,2k	9k	5,5k	9,8k	11,5k	13k	86k
3000BC	22k	21k	8,8k	13k	7,8k	14k	16k	19,4k	122k
2000BC	30k	32k	14k	20k	12k	22k	23k	29k	182k
1000BC	45k	48k	22k	31k	19k	34k	34k	45k	278k
500BC	68k	75k	35k	48k	30k	52k	50k	72k	430k
1AD	95k	105k	50k	68k	44k	74k	70k	104k	610k
500AD	145k	160k	75k	100k	68k	110k	102k	160k	920k
800AD	220k	240k	110k	150k	105k	165k	145k	255k	1,39M
900AD	285k	310k	145k	195k	145k	220k	180k	340k	1,82M
1000AD	380k	430k	200k	270k	210k	310k	230k	480k	2,51M
1100AD	490k	620k	280k	380k	310k	430k	290k	650k	3,45M
1200AD	620k	920k	390k	520k	430k	580k	360k	850k	4,67M
1250AD	590k	740k	330k	370k	390k	430k	340k	660k	3,85M
1300AD	630k	860k	440k	410k	370k	460k	380k	780k	4,33M
1350AD	670k	980k	590k	440k	380k	490k	430k	860k	4,84M
1400AD	610k	940k	620k	390k	340k	450k	450k	690k	4,49M
1450AD	680k	1,18M	830k	460k	380k	530k	520k	760k	5,34M
1500AD	810k	1,49M	1,08M	560k	430k	650k	610k	860k	6,49M
1550AD	940k	1,78M	1,31M	690k	480k	820k	710k	920k	7,65M
1600AD	760k	1,58M	1,12M	620k	410k	740k	680k	840k	6,75M
1650AD	880k	1,98M	1,39M	810k	460k	980k	790k	990k	8,28M
1700AD	1,01M	2,45M	1,69M	1,05M	540k	1,35M	915k	1,18M	10,19M
1750AD	1,38M	2,78M	1,96M	1,48M	740k	1,92M	1,45M	1,64M	13,35M
1800AD	1,95M	3,35M	2,48M	2,25M	980k	2,95M	2,42M	2,47M	18,85M
1850AD	2,82M	4,12M	3,18M	3,24M	1,18M	4,15M	4,18M	3,58M	26,45M

1900AD	4,95M	5,65M	4,42M	4,85M	1,68M	6,45M	7,65M	5,85M	41,5M

Table 3. Population totals across the eight divisions of the Russian Core at 29 benchmark dates, tracing the post-glacial peopling of the forest belt, the 1250AD Mongol trough, the 1600AD Time of Troubles dip and the nineteenth-century surge in the black-earth and Ural provinces.

Appanage Population Estimates

To show how the eight broader divisions map onto the fourteenth- and fifteenth-century appanage principalities (*udelnye knyazhestva*) and Golden Horde *uluses*, Table 4 gives population and core density figures at four medieval benchmarks (1200, 1300, 1380 and 1450AD).

Appanage Principality or Ulus	1200AD	1300AD	1380AD	1450AD	Core Density (inh./km², 1380)
Novgorod Republic	510k	505k	525k	530k	8,2 (Ilmen-Volkhov)
Pskov Republic	110k	125k	140k	150k	6,4 (Velikaya basin)
Vladimir, Pereyaslavl, Yuriev & Starodub	480k	410k	450k	540k	21,5 (Suzdal Opolye)
Rostov, Yaroslavl, Uglich, Mologa & Kostroma	330k	325k	375k	460k	9,8 (Rostov-Nero)
Beloozero & Vologda-Sheksna Appanages	110k	125k	145k	180k	2,4 (Sheksna-Beloye)
Grand Principality of Moscow, Mozhaysk, Vereya & Serpukhov	210k	240k	360k	545k	14,2 (Moskva-Radonezh)
Grand Principality of Tver, Kashin, Mikulin & Rzhev	180k	200k	245k	285k	11 (Upper Volga-Tvertsa)
Grand Principality of Ryazan, Pronsk & Yelets	410k	320k	335k	365k	7,8 (Middle Oka-Pronya)
Principality of Murom & Meshchera	110k	90k	90k	95k	4,5 (Lower Oka)
Principality of Smolensk, Vyazma & Toropets	430k	370k	365k	380k	5,2 (Upper Dnieper)
Principality of Bryansk, Karachev & Kursk-Posemye	340k	250k	255k	280k	5,8 (Desna-Sudost)
Upper Oka Appanages (Novosil, Odoev, Vorotynsk, Belyov, Tarusa, Mezetsk)	240k	210k	225k	250k	7,5 (Upper Oka-Upa-Zusha)
Nizhny Novgorod-Suzdal & Gorodets Principality	190k	205k	235k	270k	8,4 (Volga-Oka confluence)
Vyatka Land (Khlynov, Kotelnich, Orlov, Karino)	95k	100k	120k	155k	2,8 (Middle Vyatka)
Great Perm Principality (Cherdyn, Pokcha, Upper Kama)	75k	75k	85k	95k	1,1 (Upper Kama-Vishera)

Volga Bulgaria/Bulgar Ulus/Kazan Khanate	680k	560k	570k	580k	14,5 (Bolgar-Kazan core)
Ulus of Mukhshi (Narovchat), Kasimov & Temnikov Principality	170k	220k	220k	180k	6,2 (Moksha-Tsna-Oka)

Table 4. Breakdown of the eight regions into 17 individual appanage principalities and Golden Horde uluses at four medieval benchmarks, showing the rapid rise of Moscow (from 210k in 1200AD to 545k in 1450AD) and Tver alongside the slower recovery of the southern borderlands.

Regional Density Estimates

Table 5 sets out the main natural and farming settlement zones across the Russian Core and the Middle Volga-Ural basin from 10000BC to 1900AD, along with their local densities and main sources.

Settlement Zone	Date Range	Density (inh./km^2)	Area (km^2)	Geography & Historical/ Archaeological Basis
Butovo and Veretye Mesolithic	10000 - 5500BC	0,012	180k	Lacha-Vozhe-Sukhona & Volga-Oka peat-bog lake narrows (Nizhneye Veretye I, Stanovoye 4; Oshibkina 2006; Zhilin 2019, 2022)
Lyalovo and Volosovo Neolithic	5500 - 2500BC	0,065	240k	Upper/Middle Volga, Oka, Klyazma & Kama multi-hearth longhouse villages (60-200 m^2, 15-40 persons/ house; Sakhtysh I-VIII; Kostyleva et al. 2014, 2022)
Fatyanovo and Abashevo Bronze Age	2800 - 1500BC	0,18	310k	Upper Volga-Moskva-Oka-Sura-Sviyaga herding and farming territories (>300 cemeteries led by Volosovo-Danilovsky; Nordqvist & Heyd 2020; Saag et al. 2021)
Dyakovo and Gorodets Iron Age	800BC - 750AD	0,55	420k	Moskva, Upper Volga, Oka, Desna & Kama bluff hillfort chains (0,15-0,5 ha gorodishcha every 3-7 km, 50-150 persons/fort; Krenke 2011; Lopatina 2023)
Ilmen-Volkhov Poozerye Core	800 - 1900AD	14	6,5k	Lake Ilmen Poozerye, upper Volkhov (Rurikovo Gorodishche, Veliky Novgorod) & lower Shelona/Msta/ Lovat pogost core (Nosov 1990; Shapiro et al. 1971)
Suzdal Opolye Loess Core	1050 - 1900AD	23	4,5k	Suzdal-Yuriev-Pereyaslavl-Vladimir dark-soil Opolye loess islands (400 medieval selishcha, 25-30% covering 5-19 ha; Modin et al. 2020; Shpolyanskiy 2021)

Moscow-Tver Watershed Hills	1280 - 1900AD	18,5	18k	Moscow-Radonezh-Zvenigorod-Dmitrov-Tver morainic watershed hills (three-field hamlet clearing & Zamoskovny cadastres; Chernov 1991; Boytsov et al. 2020)
Upper Don Forest-Steppe	1150 - 1900AD	44,8	105k	Ryazan, Pronsk & Upper Don forest-steppe (medieval selishcha followed by 1719-1897 Tambov/ Voronezh/Ryazan farming boom; Vorotyntsev 2016; Myakisheva 2018)
Bilyar-Bolgar Cheremshan Core	950 - 1400AD	18	16,5k	Western Zakamye (Bilyar 800 ha, Bolgar 380 ha, Suvar 95 ha, Cheremshan-Aktay black-earth farming heartland; Khuzin 2001; Rudenko 2012, 2018)
Kazan Predkamye Core	1380 - 1900AD	32,4	175k	Middle Volga (Kazan, Simbirsk, Penza & Kasimov/Temnikov from 1565-1568 cadastres to 1897 Census; Mustafina 2008; Abdullin et al. 2013; Troynitsky 1905)
Central Black Earth Belt	1719 - 1900AD	58,2	108k	Kursk, Orel, Tula & Kaluga forest-steppe and black-earth guberniyas from the 1st-10th Revisions to the 1897 Census (Kabuzan 1971; Rashin 1956; Troynitsky 1905)
Moscow and St. Petersburg Industrial	1861 - 1900AD	68,5	65k	Moscow & Saint Petersburg industrial districts in the 1897 First All-Russian Imperial Census (Troitskaya & Ulyanova 2017; Rashin 1956; Troynitsky 1905)

Table 5. Local settlement zones used to scale population across the Russian Core, comparing inhabited area (km^2) and typical density (inhabitants/km^2) from Mesolithic lakeside camps (0,012/km^2) and the medieval Suzdal Opolye (23/km^2) to the nineteenth-century Central Black Earth belt (58,2/km^2).

Sources

1. Vodarsky, Yaroslav Yevgenyevich. *Naseleniye Rossii Za 400 Let (XVI - Nachalo XX Vv.) [the Population of Russia over 400 Years]*. Prosveshcheniye, 1973, https://search.worldcat.org/search?q=Population+Russia+over+400+Years+Vodarsky+1973.
2. Kuza, Andrey Vasilyevich. *Malyye Goroda Drevney Rusi [Small Towns of Ancient Rus]*. Nauka, 1989, https://search.worldcat.org/search?q=Small+Towns+Ancient+Rus+Kuza+1989.
3. Avdeev, Aleksandr Aleksandrovich, and Irina Alekseyevna Troitskaya. “Dorevizskiy Uchet Naseleniya v Rossii (XIII-XVII Veka): Istoriografiya, Diskussiya o Dostovernosti Informatsii [Pre-Revision Population Accounting in Russia (13th-17th Centuries)].” *Demograficheskoye Obozreniye / Demographic Review*, vol. 12, no. 1, 2025, pp. 4-33, https://doi.org/10.17323/demreview.v12i1.26573.

4. Oshibkina, Svetlana Viktorovna. *Mezolit Vostochnogo Prionezhya: Kultura Veretye [the Mesolithic of the Eastern Onega Region: The Veretye Culture]*. Nauka, 2006, https://search.worldcat.org/search?q=Mesolithic+Eastern+Onega+Region+Veretye+Culture+Oshibkina+2006.
5. Zhilin, Mikhail Gennadievich. "Podveski Iz Kosti v Mezolite Lesnoy Zony Vostochnoy Yevropy [Bone Pendants in the Mesolithic of the Forest Zone of Eastern Europe]." *Arkheologiya Yevraziyskikh Stepey*, no. 4, 2019, pp. 278-90, https://cyberleninka.ru/article/n/podveski-iz-kosti-v-mezolite-lesnoy-zony-vostochnoy-evropy.
6. Zhilin, Mikhail Gennadievich. "Shirokiye Nozhi s Ornamentom i Izobrazheniyami v Mezolite Volgo-Okskogo Mezhdurechya [Wide Knives with Ornament and Images in the Mesolithic of the Volga-Oka Interfluve]." *Camera Praehistorica*, no. 1, 2022, pp. 44-61, https://doi.org/10.31250/2658-3828-2022-1-.
7. Kostyleva, Elena Leonidovna, et al. "Importnyye Orudiya Russko-Karelskogo Tipa Iz Raskopok Sakhtyshskikh Stoyanok: Stratigrafiya, Planigrafiya, Khronologiya [Imported Tools of the Russo-Karelian Type from the Sakhtysh Sites]." *Vestnik Ivanovskogo Gosudarstvennogo Universiteta. Seriya: Gumanitarnyye Nauki*, no. 4, 2014, pp. 34-41, https://cyberleninka.ru/article/n/importnye-orudiya-russko-karelskogo-tipa-iz-raskopok-sahtyshskih-stoyanok-stratigrafiya-planigrafiya-hronologiya.
8. Kostyleva, Elena Leonidovna, et al. "Kostyanyye Figurki Ptits Volosovskoy Kultury: Kontekst Mestonakhozhdeniya (Po Materialam Sakhtyshskikh Stoyanok v Verkhnem Povolzhye) [Bone Bird Figurines of the Volosovo Culture at Sakhtysh]." *Samarskiy Nauchnyy Vestnik*, vol. 11, no. 1, 2022, pp. 178-87, https://doi.org/10.55355/snv2022111205.
9. Nordqvist, Kerkko, and Volker Heyd. "The Forgotten Child of the Wider Corded Ware Family: Russian Fatyanovo Culture in Context." *Praehistorische Zeitschrift*, vol. 95, no. 1, 2020, pp. 65-95, https://doi.org/10.1515/pz-2020-0002.
10. Saag, Lehti, et al. "Genetic Ancestry Changes in Stone to Bronze Age Transition in the East European Plain." *Science Advances*, vol. 7, no. 4, 2021, p. eabd6535, https://doi.org/10.1126/sciadv.abd6535.
11. Engovatova, Asya Viktorovna, et al. "Novyye Dannyye Yestestvennonauchnykh Issledovaniy Materialov Starshego Nikitinskogo Mogilnika i Yego Mesto v Khronologii Srednevolzhskoy Abashevskoy Kultury." *Samarskiy Nauchnyy Vestnik*, vol. 10, no. 3, 2021, pp. 148-52, https://doi.org/10.17816/snv2021103203.
12. Krenke, Nikolay Aleksandrovich. *Dyakovo Gorodishche: Kultura Naseleniya Basseyna Moskvy-Reki v i Tys. Do n.e. - i Tys. N.e. [Dyakovo Hillfort: Culture of the Population of the Moskva River Basin]*. Nauka, 2011, https://search.worldcat.org/search?q=Dyakovo+Hillfort+Culture+Population+Moskva+River+Basin+Krenke+2011.
13. Krenke, Nikolay Aleksandrovich. *Drevnosti Basseyna Moskvy-Reki Ot Neolita Do Srednevekovya [Antiquities of the Moskva River Basin from the Neolithic to the Middle Ages]*. Smolensk, 2019, https://search.worldcat.org/search?q=Antiquities+Moskva+River+Basin+Neolithic+Middle+Ages+Krenke+2019.
14. Lopatina, Olga Anatolevna. "Gipoteza v. A. Gorodtsova Ob Odnokrylykh Strelakh Kashirskogo Tipa: Novyy Vzglyad Na Problemu [Single-Winged Kashira-Type Arrowheads of Iron Age Dyakovo Hillforts]." *Istoricheskiye Issledovaniya*, vol. 18, 2023, pp. 25-41, https://cyberleninka.ru/article/n/gipoteza-v-a-gorodtsova-ob-odnokrylyh-strelah-kashirskogo-tipa-novyy-vzglyad-na-problemu.
15. Tikhomirov, Mikhail Nikolayevich. *Drevnerusskiye Goroda [Old Russian Towns]*. Gosudarstvennoye izdatelstvo politicheskoy literatury, 1956, https://search.worldcat.org/search?q=Old+Russian+Towns+Tikhomirov+1956.
16. Kuza, Andrey Vasilyevich. *Drevnerusskiye Gorodishcha x-XIII Vv.: Svod Arkheologicheskikh Pamyatnikov [Old Russian Fortified Settlements of the 10th-13th Centuries]*. Khristianskoye izdatelstvo, 1996, https://se

arch.worldcat.org/search?q=Old+Russian+Fortified+Settlements+10th+13th+Centuries+Kuza+1996.

17. Darkevich, Vladislav Petrovich, and G. V. Borisevich. *Drevnyaya Stolitsa Ryazanskoy Zemli: XI-XIII Vv. [The Ancient Capital of the Ryazan Land: 11th-13th Centuries]*. Krug, 1995, https://search.worldcat.org/search?q=Ancient+Capital+Ryazan+Land+11th+13th+Centuries+Darkevich+1995.
18. Yanin, Valentin Lavrentyevich. *Novgorodskaya Feodalnaya Votchina [the Novgorodian Feudal Estate]*. Nauka, 1982, https://search.worldcat.org/search?q=Novgorodian+Feudal+Estate+Yanin+1982.
19. Khuzin, Fayaz Sharipovich. *Bulgarskiy Gorod v x - Nachale XIII Vv. [The Bulgar City in the 10th - Early 13th Centuries]*. Master-Lain, 2001, https://search.worldcat.org/search?q=Bulgar+City+10th+Early+13th+Centuries+Khuzin+2001.
20. Modin, I. N., et al. "Geofizicheskiye Issledovaniya Ne Vyrazhennogo Na Poverkhnosti Srednevekovogo Nekropolya Shekshovo-9 (Suzdalskoye Opolye)." *Vestnik Moskovskogo Universiteta. Seriya 4. Geologiya*, no. 6, 2020, pp. 3-15, https://cyberleninka.ru/article/n/geofizicheskie-issledovaniya-ne-vyrazhennogo-na-poverhnosti-srednevekovogo-nekropolya-shekshovo-9-suzdalskoe-opolie.
21. Shpolyanskiy, Sergey Vladimirovich. "Sela Suzdalskogo Opolya i Izmeneniya v Sisteme Rasseleniya Vo Vtoroy Polovine XIV-XV Vv. [Villages of the Suzdal Opolye and Changes in the Settlement System in the Late 14th-15th Centuries]." *Povolzhskaya Arkheologiya*, vol. 36, no. 2, 2021, pp. 148-63, https://doi.org/10.24852/pa2021.2.36.148.163.
22. Lapshin, Vladimir Anatolevich, and Valeriy Nikolayevich Sedykh. "Osnovnyye Tendentsii v Arkheologicheskom Izuchenii Severo-Vostochnoy Rusi Na Sovremennom Etape." *Vestnik Sankt-Peterburgskogo Universiteta. Istoriya*, no. 1, 2011, pp. 39-46, https://cyberleninka.ru/article/n/osnovnye-tendentsi-v-arheologicheskom-izuchenii-severo-vostochnoy-rusi-na-sovremennom-etape.
23. Nosov, Yevgeniy Nikolayevich. *Novgorodskoye (Ryurikovo) Gorodishche [the Novgorod (Rurik) Hillfort]*. Nauka, 1990, https://search.worldcat.org/search?q=Novgorod+Rurik+Hillfort+Nosov+1990.
24. Chernov, Sergey Zarembo. "Arkheologicheskiye Dannyye o Vnutrenney Kolonizatsii Moskovskogo Knyazhestva XIII-XV Vv. I Proiskhozhdeniye Volostnoy Obshchiny." *Sovetskaya Arkheologiya*, no. 4, 1991, pp. 112-33, https://cyberleninka.ru/article/n/arheologicheskiye-dannyye-o-vnutrenney-kolonizatsii-moskovskogo-knyazhestva-xiii-xv-vv-i-proishozhdeniye-volostnoy-obschiny.
25. Boytsov, Igor Aleksandrovich, et al. "Arkheologicheskiye, Geologo-Geomorfologicheskiye i Arkhivnyye Istochniki Virtualnoy Rekonstruktsii Istoricheskogo Landshafta Vostochnoy Chasti Belogo Goroda Moskvy (XIV-XVI Vv.)." *Istoricheskaya Informatika*, no. 4, 2020, pp. 93-149, https://doi.org/10.7256/2585-7797.2020.4.34625.
26. Vorotyntsev, Leonid Vyacheslavovich. "Verkhniy Don Na Torgovykh Putyakh Zolotoy Ordy [the Upper Don on the Trade Routes of the Golden Horde]." *Istoriya: Fakty i Simvoly*, no. 3, 2016, pp. 51-58, https://cyberleninka.ru/article/n/verhniy-don-na-torgovyh-putyah-zolotoy-ordy.
27. Rudenko, Konstantin Aleksandrovich. "Vozniknoveniye Gorodov v Volzhskoy Bulgarii i Bulgarskom Uluse Zolotoy Ordy (Po Arkheologicheskim Dannym)." *Povolzhskaya Arkheologiya*, no. 1, 2012, pp. 68-77, https://doi.org/10.24852/pa2012.1.1.68.77.
28. Rudenko, Konstantin Aleksandrovich. "On the Relations of Volga Bulgaria and the Bulgar Ulus of the Golden Horde with Iran in the 10th-14th Centuries." *Zolotoordynskoye Obozreniye / Golden Horde Review*, vol. 6, no. 3, 2018, pp. 472-88, https://doi.org/10.22378/2313-6197.2018-6-3.472-488.
29. Allsen, Thomas T. *Mongol Imperialism: The Policies of the Grand Qan möngke in China, Russia, and the Islamic Lands, 1251-1259*. University of California Press, 1987, https://search.worldcat.org/search?q=Mongol+Imperialism+Policies+Grand+Qan+ongke+China+Russia+Allsen+1987.
30. Vernadsky, George. *The Mongols and Russia*. Yale University Press, 1953, https://search.worldcat.org/search?q=Mongols+Russia+Vernadsky+1953.
31. Gorsky, Anton Anatolevich. "Utverzhdeniye Vlasti Mongolskoy Imperii Nad Rusyu: Regionalnyye

Osobennosti [Consolidation of Mongol Power over Rus and the 1257-1259 Chislo]." *Istoricheskiy Vestnik*, vol. 10, no. 157, 2014, pp. 58-79, https://cyberleninka.ru/article/n/utverzhdenie-vlasti-mongolskoy-imperii-nad-rusyu-regionalnye-osobennosti.

32. Shapiro, Aleksandr Lvovich, et al. *Agrarnaya Istoriya Severo-Zapada Rossii: Vtoraya Polovina XV - Nachalo XVI v. [Agrarian History of North-Western Russia: Late 15th - Early 16th Century]*. Nauka, 1971, https://search.worldcat.org/search?q=Agrarian+History+North+Western+Russia+Late+15th+Early+Shapiro+1971.
33. Kopanev, Aleksandr Ilyich. "Naseleniye Russkogo Gosudarstva v XVI v. [Population of the Russian State in the 16th Century]." *Istoricheskiye Zapiski*, vol. 64, 1959, pp. 233-54, https://cyberleninka.ru/article/n/naseleniye-russkogo-gosudarstva-v-xvi-v.
34. Mustafina, Dina Abdulbarovna. "Materialy Opisaniy 1685-1687 Gg. Kak Istochnik Po Istorii Zemlevladeniya v Kazanskom Uezde." *Uchenyye Zapiski Kazanskogo Universiteta. Seriya Gumanitarnyye Nauki*, vol. 150, no. 1, 2008, pp. 89-96, https://cyberleninka.ru/article/n/materialy-opisaniy-1685-1687-gg-kak-istochnik-po-istorii-zemlevladeniya-v-kazanskom-uezde.
35. Vodarsky, Yaroslav Yevgenyevich. *Naseleniye Rossii v Kontse XVII - Nachale XVIII Veka (Chislennost, Soslovno-Klassovyy Sostav, Razmeshcheniye) [the Population of Russia in the Late 17th - Early 18th Century]*. Nauka, 1977, https://doi.org/10.2307/1853555.
36. Kabuzan, Vladimir Maksimovich. *Narodonaseleniye Rossii v XVIII - Pervoy Polovine XIX v. (Po Materialam Reviziy) [the Population of Russia in the 18th - First Half of the 19th Century According to the Revisions]*. Izdatelstvo Akademii nauk SSSR, 1963, https://search.worldcat.org/search?q=Population+Russia+18th+First+Half+19th+Century+according+Kabuzan+1963.
37. Kabuzan, Vladimir Maksimovich. *Izmeneniya v Razmeshchenii Naseleniya Rossii v XVIII - Pervoy Polovine XIX v. (Po Materialam Reviziy) [Changes in the Distribution of the Population of Russia in the 18th - First Half of the 19th Century]*. Nauka, 1971, https://search.worldcat.org/search?q=Changes+Distribution+Population+Russia+18th+First+Half+19th+Kabuzan+1971.
38. Ulyanova, Galina Nikolayevna, and Irina Alekseyevna Troitskaya. "Revizskiye Skazki Kak Istochnik v Istoriografii 1970-Kh - 2010-Kh Godov [Poll-Tax Revision Lists as a Source in the Historiography of the 1970s-2010s]." *Vestnik PSTGU. Seriya II: Istoriya*, vol. 71, no. 4, 2016, pp. 105-20, https://cyberleninka.ru/article/n/revizskie-skazki-kak-istochnik-v-istoriografii-1970-h-2010-h-godov.
39. Troitskaya, Irina Alekseyevna, and Galina Nikolayevna Ulyanova. "Sotsialnaya i Territorialnaya Mobilnost Moskovskikh Kuptsov Po Skazkam VIII (1833-1834) i x (1857-1858) Reviziy [Social and Territorial Mobility in Moscow According to the 8th and 10th Revisions]." *Vestnik PSTGU. Seriya II: Istoriya*, vol. 78, 2017, pp. 41-56, https://cyberleninka.ru/article/n/sotsialnaya-i-territorialnaya-mobilnost-moskovskih-kuptsov-po-skazkam-viii-1833-1834-i-x-1857-1858-reviziy.
40. Rashin, Adolf Grigoryevich. *Naseleniye Rossii Za 100 Let (1811-1913 Gg.): Statisticheskiye Ocherki [the Population of Russia over 100 Years (1811-1913): Statistical Essays]*. Gosudarstvennoye statisticheskoye izdatelstvo, 1956, https://search.worldcat.org/search?q=Population+Russia+over+100+Years+1811+1913+Statistical+Rashin+1956.
41. Mironov, Boris Nikolayevich. *Sotsialnaya Istoriya Rossii Perioda Imperii (XVIII - Nachalo XX v.): Genezis Lichnosti, Demokraticheskoy Semyi, Grazhdanskogo Obshchestva i Pravovogo Gosudarstva [the Social History of Imperial Russia]*. vol. 1, Dmitriy Bulanin, 2000, https://doi.org/10.4324/9780429499456.
42. Myakisheva, Yekaterina Aleksandrovna. "Izmeneniya Chislennosti i Soslovnogo Sostava Naseleniya Guberniy i Uezdov Yuga Tsentralnoy Rossii v 1810-1830-e Gg. [Changes in the Population Size and Estate Composition of the Guberniyas and Uezds of Southern Central Russia in the 1810s-1830s]." *Vestnik Tambovskogo Universiteta. Seriya: Gumanitarnyye Nauki*, vol. 21, no. 10, 2016, pp. 92-99, https://doi.org/10.20310/1810-0201-2016-21-12(164)-107-114.

43. Myakisheva, Yekaterina Aleksandrovna. “Osobennosti Sotsialno-Demograficheskikh Pokazateley i Professionalnogo Sostava Tambovskoy, Voronezhskoy i Ryazanskoy Guberniy v Seredine XIX Veka [Socio-Demographic Indicators of Tambov, Voronezh and Ryazan Guberniyas in the Mid-19th Century].” *Vestnik Tambovskogo Universiteta. Seriya: Gumanitarnyye Nauki*, vol. 23, no. 174, 2018, pp. 152-60, https://doi.org/10.20310/1810-0201-2018-23-1(171)-124-135.
44. Troynitsky, Nikolay Aleksandrovich. *Obshchiy Svod Po Imperii Rezultatov Razrabotki Dannykh Pervoy Vseobshchey Perepisi Naseleniya, Proizvedennoy 28 Yanvarya 1897 Goda [General Summary Across the Empire of the Results of the First General Census of Population of 28 January 1897]*. vol. 1-2, Tsentralnyy statisticheskiy komitet MVD, 1905, https://www.demoscope.ru/weekly/ssp/rus1897_01.php.
45. Akhtyamova, Alsu Vazikhovna. “Sotsialnaya Struktura Tatar v Gorodakh Ufimskoy Gubernii (Po Materialam Pervoy Vseobshchey Perepisi Naseleniya 1897 g.) [Urban Social Structure in the Volga-Ural Region According to the 1897 Census].” *Iz Istorii i Kultury Narodov Srednego Povolzhya*, no. 6, 2016, pp. 88-99, https://cyberleninka.ru/article/n/sotsialnaya-struktura-tatar-v-gorodah-ufimskoy-gubernii-po-materialam-pervoy-vseobschey-perepisi-naseleniya-1897-g.
46. Novikov, Aleksandr Viktorovich. “Keramika Poseleniya Govyadinovo (Iz Issledovaniy v. I. Smirnova, 1925-1926, 1928 Gg.) [Ceramics of the Govyadinovo Settlement in the Kostroma Volga Region].” *Arkheologiya Yevraziyskikh Stepey*, no. 4, 2019, pp. 291-316, https://cyberleninka.ru/article/n/keramika-poseleniya-govyadinovo-iz-issledovaniy-v-i-smirnova-1925-1926-1928-gg.
47. Belavin, Andrey Mikhaylovich. “Arkheologicheskiye Issledovaniya IGI UrO RAN [Archaeological Researches of Cis-Ural and Great Perm Settlements].” *Vestnik Permskogo Federalnogo Issledovatelskogo Tsentra*, no. 2, 2023, pp. 6-19, https://doi.org/10.7242/2658-705X/2023.3.6.
48. Sevostyanov, Aleksey Sergeevich. “Issledovaniye Arkheologicheskoy Keramiki Iz Selishcha Maksimkovo: Problema Tipologizatsii i Identifikatsii Ornamentov.” *Kulturnoye Naslediye Rossii*, no. 2, 2022, pp. 58-65, https://doi.org/10.34685/HI.2022.37.2.007.
49. Medovichev, A. E. “Rus v IX-x Vekakh: Arkheologicheskaya Panorama [Rus in the 9th-10th Centuries: Archaeological Panorama].” *Sotsialnyye i Gumanitarnyye Nauki. Seriya 5. Istoriya*, no. 1, 2014, pp. 137-50, https://cyberleninka.ru/article/n/2014-01-037-rus-v-ix-x-vekah-arheologicheskaya-panorama-gaydukov-p-g-gippius-a-a-grigoriev-a-v-i-dr-otv-red-makarov-n-a-ran-in-t-arheologii-m-vologda.
50. Mayorov, Anatoliy Aleksandrovich. “Drevniye Vyatichi i Ikh Osoboye Mesto Sredi Vostochnoslavyanskikh Plemen IX-XI Vv.” *Vestnik Kostromskogo Gosudarstvennogo Universiteta*, no. 6, 2015, pp. 27-31, https://cyberleninka.ru/article/n/drevnie-vyatichi-i-ih-osoboe-mesto-sredi-vostochnoslavyanskih-plemen-ix-xi-vv.
51. Kolosova, Irina Olegovna. “Monastyrskaya Tserkov Voskreseniya so Stadishcha v Batkove Ulitse Na Zapskovye. Istoriko-Topograficheskiy Kontekst.” *Pskov. Nauchno-Prakticheskiy, Istoriko-Krayevedcheskiy Zhurnal*, vol. 27, 2007, pp. 45-52, https://cyberleninka.ru/article/n/monastyrskaya-tserkov-voskreseniya-so-stadischa-v-batkove-ulitse-na-zapskovie-istoriko-topograficheskiy-kontekst-arhitekturnogo.
52. Postnikov, A. B. “Posadskiye Lyudi Pskova v Istoriografii [the Posad Population of Pskov in Historiography and Cadastral Sources].” *Pskov. Nauchno-Prakticheskiy, Istoriko-Krayevedcheskiy Zhurnal*, vol. 54, 2021, pp. 18-44, https://cyberleninka.ru/article/n/posadskie-lyudi-pskova-v-istoriografii.
53. Karpov, A. V. “Finno-Ugry i Slavyane v Volgo-Klyazminskom Mezhdurechye: Epokha Khristianizatsii [Finno-Ugrians and Slavs in the Volga-Klyazma Interfluve].” *Mir Sovremennoy Nauki*, no. 6, 2014, pp. 16-37, https://cyberleninka.ru/article/n/finno-ugry-i-slavyane-v-volgo-klyazminskom-mezhdurechie-epoha-hristianizatsii.
54. Kuchkin, Vladimir Andreyevich. *Formirovaniye Gosudarstvennoy Territorii Severo-Vostochnoy Rusi v x-XIV Vv. [Formation of the State Territory of North-Eastern Rus in the 10th-14th Centuries]*. Nauka, 1984, https://search.worldcat.org/search?q=Formation+State+Territory+North+Eastern+Rus+10th+14th+Kuchkin+1984.

55. Zhilina, Natalya Viktorovna. “Svyaz Khristianskikh Iskusstv Vo Vtoroy Polovine XI - Pervoy Polovine XII v. V Vizantii i Na Rusi.” *Problemy Istorii, Filologii, Kultury*, no. 4, 2023, pp. 178-202, https://doi.org/10.18503/1992-0431-2023-2-80-45.
56. Kuchkin, Vladimir Andreyevich. “Naseleniye Rusi v Kanun Batyeva Nashestviya [Population of Rus on the Eve of Batu’s Invasion].” *Obrazy Agrarnoy Rossii IX-XVIII Vv.*, Indrik, 1996, pp. 15-46, https://scholar.google.com/scholar?q=Population+Rus+Eve+Batu+Invasion+Kuchkin+1996.
57. Kudryashov, Aleksandr Valentinovich. “Issledovaniye Srednevekovykh Pogrebalnykh Pamyatnikov v Basseyne r. Sheksny v Nachale XXI v.” *Vestnik Cherepovetskogo Gosudarstvennogo Universiteta*, vol. 2, no. 3, 2012, pp. 36-40, https://cyberleninka.ru/article/n/issledovanie-srednevekovyh-pogrebalnyh-pamyatnikov-v-basseyne-r-sheksny-v-nachale-xxi-v.
58. Eniosova, Natalya Valerevna, et al. “Gnyozdovo - Rannegorodskoy Tsentr Smolenskoy Zemli [Gnezdovo - the Early Urban Centre of the Smolensk Land].” *Vestnik Moskovskogo Universiteta. Seriya 8. Istoriya*, no. 5, 2012, pp. 99-106, https://cyberleninka.ru/article/n/gnyozdovo-rannegorodskoy-tsentr-smolenskoy-zemli.
59. Dobrova, Olga Pavlovna. “Busy Iz Pogrebeniy Po Obryadu Ingumatsii Gnyozdovskogo Mogilnika [Beads from Inhumation Burials of the Gnezdovo Cemetery].” *Povolzhskaya Arkheologiya*, no. 3, 2017, pp. 184-96, https://doi.org/10.24852/pa2017.4.22.236.257.
60. Alekseev, Leonid Vasilyevich. *Smolenskaya Zemlya v IX-XIII Vv.: Ocherki Istorii Smolenshchiny i Vostochnoy Belorussii*. Nauka, 1980, https://search.worldcat.org/search?q=Smolenskaya+zemlya+IX+XIII+vv+Ocherki+istorii+Smolenshchiny+Alekseev+1980.
61. Tsybakov, D. L. “Problemy Opredeleniya Vladeteley Territorii Verkhney Oki Na Rubezhe XI-XII Stoletiy.” *Vestnik MGLU. Obshchestvennyye Nauki*, no. 2, 2022, pp. 112-18, https://doi.org/10.52070/2500-347X_2022_4_849_110.
62. Fennell, John. *The Crisis of Medieval Russia, 1200-1304*. Longman, 1983, https://doi.org/10.4324/9781315837192.
63. Mayorov, Aleksandr Vyacheslavovich. “Zhenshchina, Diplomatiya i Voyna: Russkiye Knyazya v Peregovorakh s Batu Nakanune Mongolskogo Nashestviya.” *Shagi / Steps*, vol. 7, no. 3, 2021, pp. 124-99, https://cyberleninka.ru/article/n/zhenschina-diplomatiya-i-voyna-russkie-knyazya-v-peregovorah-s-batu-nakanune-mongolskogo-nashestviya.
64. Bogdanov, Vladimir Olegovich, and Elena Viktorovna Lagutkina. “Sukhoputnyye Dorogi Uezda Rzhevy Volodimerovoy v XV-XVII Vv. Po Pismennym i Kartograficheskim Istochnikam.” *Istoriya: Fakty i Simvoly*, no. 1, 2025, pp. 35-48, https://doi.org/10.24888/2410-4205-2025-42-1-8-21.
65. Martin, Janet. *Medieval Russia, 980-1584*. Cambridge University Press, 2007, https://doi.org/10.1017/cbo9780511811074.
66. Belorybkin, Gennadiy Nikolayevich, et al. “Osnovnyye Itogi Arkheologicheskogo Izucheniya Narovchatskogo Gorodishcha (g. Mokhshi) i Yego Okrestnostey.” *Arkheologiya Yevraziyskikh Stepey*, no. 6, 2022, pp. 85-96, https://doi.org/10.24852/2587-6112.2022.6.85.96.
67. Inaldiev, Evgeniy Mikhaylovich. “Istoriya Izucheniya Pamyatnikov Arkheologii Zolotoordynskogo Perioda v Penzenskoy Oblasti [Ulus of Mukhshi].” *Arkheologiya Yevraziyskikh Stepey*, no. 2, 2024, pp. 142-57, https://doi.org/10.24852/2587-6112.2024.2.105.120.
68. Izmailov, Iskanderernerovich. “The Formation of the Tatar-Muslim Community in the Ik-Belaya Interfluve in the 13th-14th Centuries.” *Zolotoordynskoye Obozreniye / Golden Horde Review*, vol. 12, no. 1, 2024, pp. 186-200, https://doi.org/10.22378/2313-6197.2024-12-1.186-200.
69. Abdullin, Khalim Minnullovich, et al. “Formirovaniye Bazy Dannykh Istoricheskikh Istochnikov Vtoroy Poloviny XVI - Pervoy Poloviny XIX Veka Dlya Geoinformatsionnoy Sistemy Kazanskogo Kraya.” *Uchenyye Zapiski Kazanskogo Universiteta. Seriya Gumanitarnyye Nauki*, vol. 155, no. 3, 2013, pp.

66-78, https://cyberleninka.ru/article/n/formirovanie-bazy-dannyh-istoricheskih-istochnikov-vtoroy-poloviny-xvi-pervoy-poloviny-xix-veka-dlya-geoinformatsionnoy-sistemy.

70. Belyakov, Andrey Vasilyevich. “Nevidimki Russkoy Armii XVI Veka [Service Tatars of Kasimov, Kadom and Temnikov in the 16th-Century Russian Army].” *Istoriya Voyennogo Dela: Issledovaniya i Istochniki*, 2013, pp. 134-56, https://cyberleninka.ru/article/n/nevidimki-russkoy-armii-xvi-veka.
71. Belyakov, Andrey Vasilyevich, and G. A. Engalycheva. “Temnikovskoye Knyazhestvo Po Istochnikam XVI-XVII Vv. [The Temnikov Principality and Mordvin Belyaki].” *Srednevekovyye Tyurko-Tatarskiye Gosudarstva*, vol. 6, 2014, pp. 63-71, https://cyberleninka.ru/article/n/temnikovskoe-knyazhestvo-po-istochnikam-xvi-xvii-vv.
72. Kryazhevskikh, Andrey Leonidovich. “Istoriya Izucheniya Arkheologicheskikh Pamyatnikov v Cherte Sovremennogo g. Kirova [Archaeological Monuments Within Medieval Khlynov].” *Vestnik Vyatskogo Gosudarstvennogo Universiteta*, no. 1, 2011, pp. 24-33, https://cyberleninka.ru/article/n/istoriya-izucheniya-arheologicheskih-pamyatnikov-v-cherte-sovremennogo-g-kirova.
73. Egorov, Andrey Vladimirovich. “Gipotezy o Vremeni Poyavleniya Karinskikh Tatar Na Territorii Vyatskogo Kraya v Svete Arkheologicheskikh Istochnikov.” *Vestnik Vyatskogo Gosudarstvennogo Universiteta*, no. 12, 2014, pp. 45-51, https://cyberleninka.ru/article/n/gipotezy-o-vremeni-poyavleniya-karinskih-tatar-na-territorii-vyatskogo-kraya-v-svete-arheologicheskih-istochnikov.
74. Krylasova, Natalya Borisovna. “Trudy Po Arkheologii Rodanovskoy i Lomovatovskoy Kultur Permi Velikoy.” *Trudy Kamskoy Arkheologo-Etnograficheskoy Ekspeditsii*, vol. 13, 2017, pp. 12-31, https://cyberleninka.ru/article/n/spisok-trudov-n-b-krylasovoy.
75. Danich, A. V. “Pronizki Piter (Stepanovo Plotbishche) Mogilnika Vtoroy Poloviny x-XI Vv.” *Trudy Kamskoy Arkheologo-Etnograficheskoy Ekspeditsii*, 2025, pp. 1-17, https://doi.org/10.24412/2658-7637-2025-27-90-106.
76. Selin, Adrian Aleksandrovich. “Izgnannyye Iz Livonii: Ob Ustroystve Pomeshchikov Starykh i Novykh Nemetskikh Gorodov Posle Plyusskogo Peremiriya.” *Studia Slavica Et Balcanica Petropolitana*, no. 1, 2020, pp. 121-32, https://doi.org/10.21638/spbu19.2020.202.